

Write the correct answers

2 - 2 - 2

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
b	d	c	c	c	d	c	d	a a	b
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
c	b	a	d	d	a	d	b	d	b

Q1) If $f(x) = \sqrt{4-x^2}$, then domain f and Range f are

- a) $D_f: (-\infty, -2] \cup [2, \infty)$, $R_f: [0, 2]$ (b) $D_f: [-2, 2]$, $R_f: [0, 2]$ c) $D_f: [-2, 2]$, $R_f: [0, \infty)$ d) $D_f: [0, 2]$, $R_f: [0, \infty)$

Q2) If $f(x) = \frac{3x+1}{x-2}$, then $f^{-1}(x) =$

- a) $\frac{2x+1}{x+3}$ b) $\frac{x-2}{3x+1}$ c) $\frac{x+1}{x-3}$ (d) $\frac{2x+1}{x-3}$

* Q3) If $f(x) = \begin{cases} [x-c^2] & , x \leq 3 \\ 2x-4 & , x > 3 \end{cases}$ is continuous at $x=3$, then the values of c

Where $[...]$ in f is the greatest integer function

- a) $[-1, 0]$ (b) $[0, 1]$ (c) $[-1, 1]$ d) $[-1, 1] \setminus \{0\}$

Q4) One of the following statements is false

- a) The domain of $f - g$ is the intersection of the domains of f and g .
 b) The domain of $g \circ f$ consists of all values in domain f and values of x where $f(x)$ is in domain g .
 (c) The graph of $f(-x)$ obtained by reflect graph f about x -axis.
 d) The graph of even functions is symmetric about y -axis.

Q5) If $g(x) = \frac{f(x) + f(-x)}{2}$, $h(x) = \frac{f(x) - f(-x)}{2}$, where f has domain of all real numbers, then

- a) g is odd and h is even function
 b) g is even and h is odd function
 c) g and h are odd functions
 d) g and h are even functions

Q6) The vertical and horizontal asymptotes for $f(x) = \frac{2x^2 - 8}{x^2 - 3x + 2}$ respectively are

- a) $x=1, 2$ $y=2$ b) $x=-1, y=2$ c) $x=2, y=1$ d) $x=1, y=2$

Q7) If $\lim_{x \rightarrow \infty} \csc\left(\frac{\pi x^2}{2 - 3x^2}\right)$

- a) $\frac{-1}{\sqrt{3}}$ b) -2 c) $-\frac{2}{\sqrt{3}}$ d) $\frac{-1}{3}$

Q8) If the graph of the function f is reflected about x -axis, stretched vertically by 2 and then shifted left by 3 units. The new function will be =

- a) $-2f(x+3)$ b) $2f(-x+3)$ c) $-\frac{1}{2}f(x+3)$ d) $2f(-x-3)$

Q9) The discontinuity points of $f(x) = \frac{|x-2|}{\csc x - 2}$

- a) $x = \frac{\pi}{6} \pm 2n\pi, \frac{5\pi}{6} \pm 2n\pi$ b) $x = \frac{\pi}{3} \pm 2n\pi, \frac{5\pi}{3} \pm 2n\pi$ c) $x = \frac{\pi}{3} \pm 2n\pi, \frac{2\pi}{3} \pm 2n\pi$ d) otherwise
 where $n=0, 1, 2, \dots$

Q10) one of the following functions has inverse on its natural domain

- a) $f(x) = x^2 - 3$ b) $f(x) = x^3 + 4$ c) $f(x) = |x - 5|$ d) $f(x) = \cos x$

Q11) $\lim_{x \rightarrow \infty} \frac{(3x-1)^2(2x^2+1)}{5x(3x+3)^3} =$

- a) 1 b) $\frac{2}{3}$ c) $\frac{2}{15}$ d) $\frac{3}{5}$

Q12) The interval of continuity for $f(x) = \sqrt{|x-4|+3x}$ is

- a) $[4, \infty)$ b) $[-2, \infty)$ c) $[2, 4]$ d) $[0, 4]$

Q13) The graph of the curve $y^3 = 2x \cos x$ is symmetric about

- a) origin b) x -axis c) y -axis d) No symmetry

Q14) If $f(x) = \csc^2 \sqrt{x}$, then

a) $D_f : x \geq 0$ and $x \neq n^2 \pi^2, R_f : (0, 1]$

b) $D_f : x \geq 0$ and $x \neq n^2 \pi^2, R_f : (0, 1]$

c) $D_f : x > 0$ and $x \neq 2n^2 \pi^2, R_f : [1, \infty)$

(d) $D_f : x > 0$ and $x \neq n^2 \pi^2, R_f : [1, \infty)$

Q15) $\lim_{x \rightarrow \infty} x \left(\sin \frac{2}{x} + \frac{1}{x^2} \cos 3x \right) =$

a) $+\infty$

b) 5

c) 3

(d) 2

Q16) $\lim_{x \rightarrow 0} \frac{\tan^2 ax - 2x^2}{4x^2 - \sin ax^2} = 1$, then $a =$

(a) -3, 2

b) 2

c) 3, -2

d) 1, -3

* Q17) $f(x) = \begin{cases} |x-5| & , x \leq 2 \\ x^2 - 9 & , 2 < x \leq 3 \\ 2x - 6 & , x > 3 \end{cases}$, $g(x) = \begin{cases} \cos x + 2 & , x \leq 0 \\ x^2 - 2x + 3 & , 0 < x < 1 \\ |x-2| + 1 & , x \geq 1 \end{cases}$

$f \circ g$ is continuous at

a) $x=0, 1$

b) $x=1$

c) $x=1, 3$

(d) $x=0$

Q18) $f(x) = x + \frac{1}{x}$, $g(x) = \frac{x-1}{x+2}$, the domain of $f \circ g$ is

a) $x=0, 1$

(b) $x=1, -2$

c) $x \neq 0, 1, -2$

d) $x \neq 0, -2$

Q19) $\lim_{x \rightarrow 3} \frac{\sqrt{x^2 - 6x + 9}}{x^2 - 6x + 9}$

a) $-\infty$

b) 0

c) -1

(d) $+\infty$

Q20) $\lim_{x \rightarrow \infty} x + \sqrt{x^2 + 4x}$

a) -1

(b) -2

c) $-\infty$

(d) $+\infty$

Best wishes



أسئلة سنوات سابقة

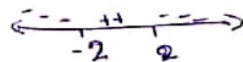
المادة:

Q₁) If $f(x) = \sqrt{4-x^2}$, then domain f and Range f are.

- a) $(-\infty, 2]$, $[2, \infty)$ $R_f: [0, 2]$ **b) $[-2, 2]$, $R_f [0, 2]$**
 c) $D_f: [-2, 2]$, $R_f [0, \infty)$ d) $D_f [0, 2]$, $R_f [0, \infty)$

Sol:-

$$\sqrt{4-x^2} = \sqrt{(2-x)(2+x)}$$



$$Df(x) = [-2, 2]$$

x	-2	0	1	2
f(x)	0	2	$\sqrt{3}$	0

 $\Rightarrow R_f = [0, 2]$

b)

Q₂) If $f(x) = \frac{3x+1}{x-2}$, then $f^{-1}(x) =$

- a) $\frac{2x+1}{x+3}$ b) $\frac{x-2}{3x+1}$ c) $\frac{x+1}{x-3}$ **d) $\frac{2x+1}{x-3}$**

Sol:-

$$y = \frac{3x+1}{x-2}$$

$$x = \frac{3y+1}{y-2} \rightarrow xy - 2x = 3y+1 \rightarrow xy - 3y = 1+2x$$

$$\rightarrow y(x-3) = 1+2x \rightarrow y = \frac{1+2x}{x-3} \rightarrow f^{-1}(x) = \frac{1+2x}{x-3}$$

d)

Q₃) If $f(x) = \begin{cases} [x - c^2], & x \leq 3 \\ 2x - 4, & x > 3 \end{cases}$

is cts at $x=3$ then the values of c where $[-\infty, \infty]$ in f is the greatest integer function

- a) $[-1, 0]$ **b) $[0, 1]$** c) $[-1, 1]$ d) $[-1, 1] \cup [0, 1]$

Don't miss the change - Be part of it



أسئلة سنوات سابقة

المادة:

Solo- Cts at $x=3 \rightarrow \lim_{x \rightarrow 3^+} 2x-4 = \lim_{x \rightarrow 3^-} [x-c^2]$

$$2 = [3-c^2] \rightarrow 2 \leq 3-c^2 \leq 3$$

$$-1 \leq -c^2 \leq 0$$

$$0 \leq c^2 \leq 1$$

$$[0, 1]$$

[b]

Q5) If $g(x) = \frac{f(x) + f(-x)}{2}$, $h(x) = \frac{f(x) - f(-x)}{2}$ where f has

domain of all real number, then.

a) g is odd and h is even function

b) g is even and h is odd

c) g and h are odd

d) g and h are even

Sol: $g(-x) = \frac{f(-x) + f(-(-x))}{2} = \frac{f(-x) + f(x)}{2} = g(x) \Rightarrow \text{even}$

$h(-x) = \frac{f(-x) - f(-(-x))}{2} = \frac{f(-x) - f(x)}{2}$

and $-h(x) = \frac{-f(x) - f(-x)}{2} = \frac{f(-x) - f(x)}{2}$

$\therefore h(-x) = -h(x) \Rightarrow \text{odd}$

(b)

Q6) The vertical and horizontal asymptotes for $f(x) = \frac{2x^2 - 8}{x^2 - 3x + 2}$ respectively are

a) $x=1, 2$ $y=2$

b) $x=1, y=2$

c) $x=2, y=1$ d) $x=1, y=2$

V.A $f(x) = \frac{2(x^2-4)}{(x-1)(x-2)} = \frac{2(x/2)(x+2)}{(x-1)(x/2)} \therefore x=1$ V.A

H.A $\lim_{x \rightarrow \infty} \frac{2x^2-8}{x^2-3x+2} = 2$ $y=2$ H.A

Don't miss the change x^2-3x+2 if

(2)



أسئلة سنوات سابقة

المادة:

Q7) If $\lim_{x \rightarrow \infty} \csc\left(\frac{\pi x^2}{x^2 - 3x + 2}\right) =$ ~~_____~~

a) $\frac{-1}{\sqrt{3}}$

b) -2

c) $-\frac{2}{\sqrt{3}}$

d) $\frac{-1}{3}$

Sol:-

$$\csc\left(\lim_{x \rightarrow \infty} \frac{\pi x^2}{x^2 - 3x + 2}\right) = \csc\left(\lim_{x \rightarrow \infty} \frac{\pi x^2}{x^2 - 3x + 2}\right) = \csc\left(-\frac{\pi}{3}\right)$$

$$= -\csc\left(\frac{\pi}{3}\right) = -\frac{2}{\sqrt{3}}$$

(c)

Q8) If the graph of the function f is reflected about x -axis stretched vertically by 2 and then shifted left by 3 units The new function will be =

(a) $-2f(x+3)$

b) $2f(-x+3)$

c) $-\frac{1}{2}f(x+3)$

d) $2f(-x-3)$

Sol:- $f(x)$

reflected x -axis $\rightarrow f(x) \rightarrow$ stretched vertically $\rightarrow -2f(x) \rightarrow$ shifted left $\rightarrow -2f(x+3)$

(a)

Q9) The discontinuity points of $f(x) = \frac{1}{\csc x - 2}$ are

(a) $x = \frac{\pi}{6} + 2n\pi, \frac{5\pi}{6} + 2n\pi$

b) $x = \frac{\pi}{3} + 2n\pi, \frac{2\pi}{3} + 2n\pi$

c) $x = \frac{\pi}{3} + 2n\pi, \frac{2\pi}{3} + 2n\pi$
where $n = 0, 1, 2, \dots$

d) otherwise

Sol:- $\csc x - 2 = 0$

$\csc x = 2$

$\sin x = \frac{1}{2} \rightarrow x = \frac{\pi}{6} + 2n\pi, x = \frac{5\pi}{6} + 2n\pi$

Don't miss the change - Be part of it

(a)



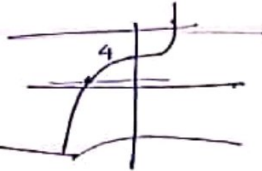
أسئلة سنوات سابقة

المادة:

Q1) one of the following functions has inverse on its natural domain.

- a) $f(x) = x^2 - 3$ **b) $f(x) = x^3 + 4$** c) $f(x) = |x - 5|$ d) $f(x) = \cos x$

Sol: **b)** $f(x) = x^3 + 4$



Q11) ~~Q11~~ $\lim_{x \rightarrow \infty} \frac{(3x-1)^2}{5x(3x+3)^3} \cdot \frac{2x^2+1}{1}$

- a) = 1 b) $\frac{2}{3}$ **c) $\frac{2}{15}$** d) $\frac{3}{5}$

Sol: $\lim_{x \rightarrow \infty} \frac{9x^2 \cdot 2x^2}{5x \cdot 27x^3} = \lim_{x \rightarrow \infty} \frac{2}{\pi \cdot 3} = \frac{2}{15}$

Q12) The interval of continuity for $f(x) = \sqrt{|x-4|+3x}$ is

- a) $[4, \infty)$ **b) $[-2, \infty)$** c) $[2, 4]$ d) $[0, 4]$

Sol: $|x-4| + 3x = 0$
 $|x-4| = -3x$
 $x-4 = -3x \rightarrow 4x = 4 \rightarrow x = 1$
 $x-4 = 3x \rightarrow 2x = -4 \rightarrow x = -2 \checkmark$



b) $[-2, \infty)$

Q13) The graph of the curve $y = 2x \cos x$ is symmetric about

- a) origin** b) x -axis c) y -axis d) No symmetry

$(-y)^3 = 2(-x) \cos(-x) \Rightarrow -y^3 = -2x \cos x$

Don't miss the change - Be part of it

a)



أسئلة سنوات سابقة

المادة:

Q.14) If $f(x) = \csc^2 \sqrt{x}$, then

a) $Df: x > 0$ and $x \neq n^2 \pi^2, R_f: (0, 1]$ b) $Df: x > 0$ and $x \neq n^2 \pi^2, R_f: [0, 1]$

✗

c) $Df: x > 0$ and $x \neq 2n^2 \pi^2, R_f: [1, \infty)$

d) $Df: x > 0$ and $x \neq n^2 \pi^2, R_f: [1, \infty)$

Sol: Domain $f(x) = \csc^2 \sqrt{x}$ then $x > 0$ and $x \neq n^2 \pi^2$

$$\text{Rang } 0 \leq \sin^2 \sqrt{x} \leq 1$$

$$1 \leq \frac{1}{\sin^2 \sqrt{x}} \leq \infty \quad [1, \infty)$$

(D)

Q.15) $\lim_{x \rightarrow \infty} \left(\sin \frac{2}{x} + \frac{1}{x^2} \cos 3x \right) =$

a) $+\infty$

b) 5

c) 3

(d) 2

Sol:-

$$-1 \leq \cos 3x \leq 1$$

$$-\frac{1}{x} \leq \frac{1}{x} \cos 3x \leq \frac{1}{x}$$

$$\lim_{x \rightarrow \infty} -\frac{1}{x} = 0$$

↓

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\frac{1}{x} = 0$$

$$\lim_{x \rightarrow \infty} \frac{\sin \frac{2}{x}}{\frac{1}{x}} + 0$$

$$2 + 0 = 2$$

(D)

Q.16) $\lim_{x \rightarrow \infty} \frac{4x^2 + ax - 2x^2}{4x^2 - \sin ax} = 1$, then $a =$

(a) -3, 2

b) 2

c) 3, 2

d) 1, -3

(a)

Sol: x^2 as $\infty \rightarrow \lim_{x \rightarrow \infty}$

$$\frac{a^2 - 2}{4 - a} = 1 \rightarrow a^2 - 2 = 4 - a \rightarrow a^2 + a - 6 = 0$$

$$(a+3)(a-2) = 0$$

$$\begin{cases} a = -3 \\ a = 2 \end{cases}$$

Don't miss the change - Be part of it

B)



أسئلة سنوات سابقة

المادة:

Q4) (C) The graph of $f(-x)$ obtained by reflect graph f about x -axis.

Q17) g cts at $0 \rightarrow g(0) = 3$ and f cts at $3 \therefore f \circ g$ cts at 0

g cts at $1 \rightarrow g(1) = 2$ and f not cts at 1

g cts at $3 \rightarrow g(3) = 2$ and f not cts at 2

$\therefore f \circ g$ not cts at 3

~~...~~

(D) $x=0$ ~~...~~

Don't miss the change ... Be part of it

Scanned by CamScanner

Scanned by CamScanner

Scanned by CamScanner



أسئلة سنوات سابقة

المادة:

Q1) $f(x) = x + \frac{1}{x}$, $g(x) = \frac{x-1}{x+2}$, the domain of $f \circ g$ is

- a) $x \neq 1$ **b) $x \neq 1, -2$** c) $x \neq 0, 1, -2$ d) $x \neq 0, 1, -2$

Sol:- $f \circ g(x) = f(g(x)) = f\left(\frac{x-1}{x+2}\right)$

$$= \frac{x-1}{x+2} + \frac{1}{\left(\frac{x-1}{x+2}\right)} = \frac{x-1}{x+2} + \frac{x+2}{x-1} = \frac{(x-1)^2 + (x+2)^2}{(x+2)(x-1)}$$

$D_{f \circ g} = D_f \cap D_g = (R - \{1, -2\}) \cap (R - \{1, -2\}) = R - \{1, -2\}$

B

Q11) $\lim_{x \rightarrow 3^-} \frac{\sqrt{x^2 - 6x + 9}}{x^2 - 6x + 9}$

a) $-\infty$

b) 0

c) -1

d) $+\infty$

Sol:- $\lim_{x \rightarrow 3^-} \frac{\sqrt{(x-3)^2}}{x^2 - 6x + 9} = \lim_{x \rightarrow 3^-} \frac{|x-3|}{(x-3)^2}$

$\lim_{x \rightarrow 3^-} \frac{-(x-3)}{(x-3)^2} = \lim_{x \rightarrow 3^-} \frac{-1}{x-3} = \frac{-1}{0^-} = \infty$

D

Q20) $\lim_{x \rightarrow -\infty} x + \sqrt{x^2 + 4x}$ Sol:- $\frac{x - \sqrt{x^2 + 4x}}{x - \sqrt{x^2 + 4x}}$

a) -1

b) -2

c) $-\infty$

d) $+\infty$

$\lim_{x \rightarrow -\infty} \frac{x^2 - (x^2 + 4x)}{x - \sqrt{x^2 + 4x}} = \lim_{x \rightarrow -\infty} \frac{x^2 - x^2}{x - \sqrt{x^2 + 4x}} = \lim_{x \rightarrow -\infty} \frac{-4x}{x - \sqrt{x^2 + 4x}}$

$= \lim_{x \rightarrow -\infty} \frac{-4x}{x - x} \rightarrow -2$

B

Don't miss the change - Be part of it

B

Instructor:

Date: 19/3/2019.

Name:

Lecture Time: ١٢:٣٠ - ١:٠٠

Circle the correct answer (20 marks)

Q1) $f(x) = \frac{x^2+1}{x+3}$, if $f^{-1}(b) = 1$, then $b =$

a) $\frac{1}{2}$

b) 1

c) $\frac{5}{3}$

d) $\frac{1}{3}$

Q2) $f(x) = \sqrt{\frac{x^2-9}{x-3}}$, $D_f =$

a) $[-3, 3)$

b) $[-3, \infty) \setminus \{3\}$

c) $(-3, 3)$

d) $(3, \infty)$

Q3) $f(x)$ is continuous on the interval $[a, b]$ if

a) f is continuous on (a, b) , $f(a) = \lim_{x \rightarrow a^+} f(x)$ and $f(b) = \lim_{x \rightarrow b^-} f(x)$

b) f is continuous on (a, b) , $f(a) = \lim_{x \rightarrow a} f(x)$ and $f(b) = \lim_{x \rightarrow b} f(x)$

c) f is continuous on (a, b) , $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow b^+} f(x) = \lim_{x \rightarrow b^-} f(x)$

d) f is continuous on (a, b) , $f(a) = \lim_{x \rightarrow a^+} f(x)$ and $f(b) = \lim_{x \rightarrow b^-} f(x)$

Q4) If graph f is shifted 2 to the left, stretched vertically by 3 and then shifted down 1 unit. The function will be

a) $3f(x+2)-1$

b) $f(3x-2)-1$

c) $f\left(\frac{1}{3}x+2\right)-1$

d) $\frac{1}{3}f(x-2)-1$

Q5) The discontinuity points for $f(x) = \frac{|x|-2}{\sec x - 2}$

a) $\frac{\pi}{3} \pm 2n\pi, \frac{5\pi}{3} \pm 2n\pi$

b) $\frac{\pi}{6} \pm 2n\pi, \frac{7\pi}{6} \pm 2n\pi$

c) $\frac{\pi}{3} \pm 2n\pi, \frac{2\pi}{3} \pm 2n\pi$

d) $\frac{\pi}{6} \pm 2n\pi, \frac{5\pi}{6} \pm 2n\pi$

Q6) $\lim_{x \rightarrow 0} \frac{\sin^3 2x}{2x^3} =$

a) 8

b) 2

c) 1

d) 4

Q7) $\lim_{x \rightarrow \infty} \frac{(2x-1)^2}{7x^2-2x+1} =$

a) $\frac{2}{7}$

b) 0

c) -1

d) $\frac{4}{7}$

Q12) (10 marks) Solve

3 a) Find vertical and horizontal asymptotes for $f(x) = \frac{2x^2 - x - 1}{x^2 - 1}$

Vertical: $x^2 - 1 = 0$

$$x = \pm 1$$

$$\lim_{x \rightarrow 1} \frac{2x^2 - x - 1}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(2x+1)}{(x-1)(x+1)} = \frac{2}{2} \text{ not vertical}$$

$$\lim_{x \rightarrow -1} \frac{(x-1)(2x+1)}{(x-1)(x+1)} = \frac{-2}{0} = \infty$$

$$x = -1$$

is vertical Asymptotes

Horizontal:-

$$\frac{2x^2}{x^2} = \frac{2}{1} = 2$$

$$y = 2$$

is horizontal Asymptotes

3 b) Write steps of translation, reflection, compress or stretch if exist for

$$\frac{1}{2} f(-x-3) + 5$$

3-1- Compress vertically by $\frac{1}{2}$

2-2- Reflect about $\frac{1}{x}$ axis

1-3- Shift Right 3 units

4-4- shift up 5 units

c) Test if the curve $y^3 = x \cos x$ is symmetric about origin or not.

$$y^3 = x \cos x$$

about origin

$$x \rightarrow -x$$

$$y \rightarrow -y$$

$$-y^3 = -x \cos -x$$

$$-y^3 = -x \cos x$$

$$y^3 = x \cos x$$

is symmetric about origin

Q8) $f(x) = 2x^2 + 4x - 1$, Range $f =$

- a) $[-3, \infty)$ b) $[3, \infty)$ c) $(-\infty, -9]$ d) $(-\infty, 7]$

Q9) $\lim_{x \rightarrow +\infty} \frac{\sin 2x}{3x} =$

- a) $\frac{2}{3}$ b) 0 c) $+\infty$ d) $-\infty$

Q10) The function $f(x) = \frac{x-3}{x^2 \cos x}$ is

- a) even function b) odd function c) even and odd function d) not even nor odd function

Write the answers

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
A	B	B	A	A	D	D	A	B	B

Q11)(10 marks) Write only the final answer

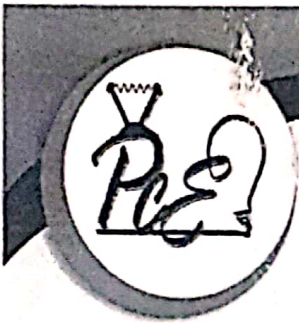
a) $\lim_{x \rightarrow -\infty} \sin\left(\frac{\sqrt{\pi^2 x^2 - 5x}}{6x}\right) = \dots\dots\dots 0 \dots\dots\dots$

b) $f(x) = \frac{\csc 3x}{|x-1|-2}$, D_f is $[3, \infty) \setminus \{3x \pm n\pi \mid n=0, 1, 2, 3, \dots\}$

c) $f(x) = 3 \sin x - 2$, $D_{f^{-1}}$ is $[-5, 1]$

d) $f(x) = \sqrt{3x-1}$, $f^{-1}(x) = \frac{x^2+1}{3}$

e) $\lim_{x \rightarrow 0^+} \frac{3}{x} - \frac{1}{x^2} = \dots\dots\dots -\infty \dots\dots\dots$



أسئلة سنوات سابقة

المادة: ميدكال كوالاس 1

Q₁) $F(x) = \frac{x^2+1}{x+3}$ $F'(b) = 1$ then $b =$

$$F'(b) = 1$$

$$b = F(1)$$

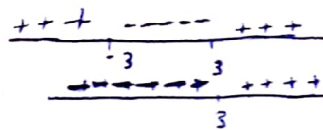
$$F(F'(b)) = F(1)$$

$$F(1) = \frac{2}{4} = \frac{1}{2}$$

Q₂) $F(x) = \sqrt{\frac{x^2-9}{x-3}}$ $D_F =$

$$x^2-9=0 \quad x = \pm 3$$

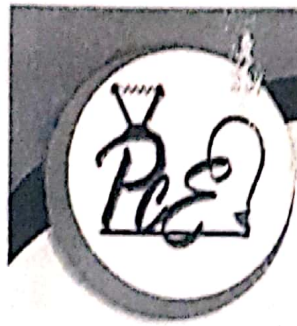
$$x-3=0 \quad x = 3$$



$D_F = [-3, \infty) \setminus \{3\}$ b

Q₃) $F(x)$ is continuous on the interval $[a, b]$ if :-

(d) F is continuous on (a, b) , $F(a) = \lim_{x \rightarrow a^+} F(x)$ and $F(b) = \lim_{x \rightarrow b^-} F(x)$



أسئلة سنوات سابقة

المادة: مبراهيم الكولاس 1

Q4) F shifted 2 to the left, stretched vertically by 3 and shifted 1 unit. The function will be-

Shift left
 $3F(x+2) - 1$ down 1 unit
stretched vertically
a.

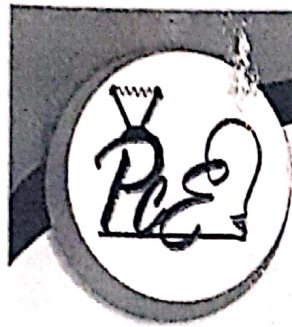
Q5) discontinuity points for $F(x) = \frac{|x|-2}{\sec x - 2}$

جد أصفاء المقام
 $\sec x - 2 = 0$
 $\sec x = 2$
 $\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}$
 $x = \frac{\pi}{3} \pm 2n\pi$
 $x = \frac{5\pi}{3} \pm 2n\pi$
a.

θ
$2\pi - \theta$

Q6) $\lim_{x \rightarrow 0} \frac{\sin^3 2x}{2x^3}$

$$\frac{\sin 2x}{2x} \cdot \frac{\sin 2x}{x} \cdot \frac{\sin 2x}{x}$$
$$\frac{2x}{2x} \cdot \frac{2x}{x} \cdot \frac{2x}{x}$$
$$1 \cdot 2 \cdot 2 = \boxed{4} \quad d.$$



أسئلة سنوات سابقة

المادة: ميد كالكولاس 1

$$Q_7) \lim_{x \rightarrow \infty} \frac{(2x-1)^2}{7x^2-2x+1}$$

$$\lim_{x \rightarrow \infty} \frac{4x^2-4x+1}{7x^2-2x+1} = \frac{4x^2}{7x^2} = \boxed{\frac{4}{7}} \quad d.$$

$$Q_8) F(x) = 2x^2 + 4x - 1$$

Range F is:-

معامل x^2 موجب

$$\left[F\left(-\frac{b}{2a}\right), \infty \right)$$

$$-\frac{b}{2a} = \frac{-4}{2 \times 2} = \frac{-4}{4} = \boxed{-1}$$

$$F(-1) = \boxed{-3}$$

$$[-3, \infty) \quad a.$$

$$Q_9) \lim_{x \rightarrow \infty} \frac{\sin 2x}{3x}$$

by squeezing theorem :-

$$-1 \leq \sin 2x \leq 1$$

$$\frac{-1}{3x} < \frac{\sin 2x}{3x} < \frac{1}{3x}$$

$$\downarrow \lim_{x \rightarrow \infty} \frac{-1}{3x}$$



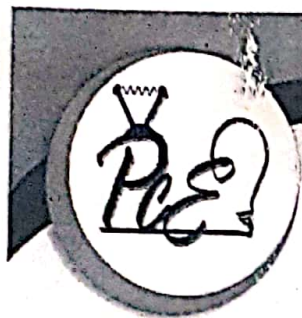
$$\downarrow \lim_{x \rightarrow \infty} \frac{1}{3x}$$

0

$$\lim_{x \rightarrow \infty} \frac{\sin 2x}{3x}$$

$$= 0$$

b.



أسئلة سنوات سابقة

المادة: ميد كالكولاس 1

Q10) The function $F(x) = \frac{x-3}{x^2 \cos x}$ is

$$F(-x) = \frac{-x-3}{(-x)^2 \cos(-x)} = \frac{-x-3}{x^2 \cos x} = \frac{-(x+3)}{x^2 \cos x} \quad \text{Not odd}$$

$$-F(x) = \frac{-x+3}{-x^2 \cos x} = \frac{-(x-3)}{x^2 \cos x} \quad \text{Not even} \quad \text{d.}$$

Q11) a) $\lim_{x \rightarrow -\infty} \sin\left(\frac{\sqrt{\pi^2 x^2 - 5x}}{\delta x}\right)$

$$\sin\left(\lim_{x \rightarrow -\infty} \frac{\sqrt{\pi^2 x^2 - 5x}}{\delta x}\right)$$

$$\sin\left(\lim_{x \rightarrow -\infty} \frac{\pi|x|}{\delta x}\right) = \sin\frac{-\pi}{\delta} = \boxed{\frac{-1}{2}}$$

b) $F(x) = \frac{\csc 3x}{|x-1|-2}$, D_F is

$$\frac{1}{\frac{\sin 3x}{|x-1|-2}}$$

$D_F: x \neq$

$$\sin 3x = 0 \Rightarrow x = \pm n\pi \rightarrow x = \pm \frac{n\pi}{3}$$

$$|x-1|-2 = 0$$

$$|x-1| = 2$$

$$x-1=2 \quad \text{or} \quad x-1=-2$$

$$x=3 \quad \text{or} \quad x=-1$$

$$D_F: x \neq \pm \frac{n\pi}{3}, 3, -1$$



أسئلة سنوات سابقة

المادة: ميثاق الكولاس 1

c) $F(x) = 3\sin x - 2$, $D_{F^{-1}}$ is:-

$$D_{F^{-1}} = R_F$$

$$R_F =$$

$$-1 \leq \sin x \leq 1$$

$$-3 \leq 3\sin x \leq 3$$

$$-5 \leq 3\sin x - 2 \leq 1$$

$$[-5, 1]$$

d) $F(x) = \sqrt{3x-1}$, $F^{-1}(x)$ is:-

$$y = \sqrt{3x-1}$$

$$x \rightarrow y$$

$$y \rightarrow x$$

$$x = \sqrt{3y-1}$$

$$x^2 = 3y-1$$

$$3y = x^2 + 1$$

$$y = \frac{x^2 + 1}{3}$$

$$F^{-1}(x) = \frac{x^2 + 1}{3}$$



أسئلة سنوات سابقة

المادة: مبداء الكولاس 1

e) $\lim_{x \rightarrow 0^+} \frac{3}{x} - \frac{1}{x^2}$

$$\lim_{x \rightarrow 0^+} \frac{3x^2 - x}{x^3}$$

$$\lim_{x \rightarrow 0^+} \frac{x(3x-1)}{x^3} \Rightarrow \lim_{x \rightarrow 0^+} \frac{3x-1}{x^2} \Rightarrow \frac{-1}{0} = \boxed{-\infty}$$

Q12) a) Find vertical and horizontal asymptotes For $f(x) = \frac{2x^2 - x - 1}{x^2 - 1}$

Vertical:-

$$x^2 - 1 = 0$$

$$x = \pm 1$$

$$\lim_{x \rightarrow 1} \frac{2x^2 - x - 1}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(2x+1)}{(x-1)(x+1)} = \frac{3}{2} \text{ not vertical}$$

$$\lim_{x \rightarrow -1} \frac{(x-1)(2x+1)}{(x-1)(x+1)} = \frac{-1}{0} = -\infty \checkmark$$

$\therefore \boxed{x = -1}$ is vertical Asymptotes

horizontal:-

$$\frac{2x^2}{x^2} = \frac{2}{1} = \boxed{2}$$

$\boxed{y = 2}$ is horizontal Asymptotes.



أسئلة سنوات سابقة

المادة: مبداء الكواليس 1

b) write steps of translation, reflection, compress or stretch
if exist for $\frac{1}{2}f(-x-3)+5$

- 1- shift right 3 units.
- 2- reflect about y-axis.
- 3- compress vertically by $\frac{1}{2}$.
- 4- shift up 5 units.

c) Test if the curve $y^3 = x \cos x$ is symmetric about origin or not:-

$$y^3 = x \cos x$$

about origin

$$x \rightarrow -x$$

$$y \rightarrow -y$$

$$-y^3 = -x \cos -x$$

$$-y^3 = -x \cos x$$

$$\boxed{y^3 = x \cos x}$$

$\therefore$ the curve is symmetric about origin.

Q1:-

$$f(x) = \ln(3 - 19x + 51)$$

sol :-

$$3 - 19x + 51 > 0 \implies \text{Domain } \ln f = f > 0$$

$$19x + 51 < 3$$

$$-3 < 9x + 5 < 3 \implies \text{خواص اللوغاريتمات}$$

$$-8 < 9x < -2$$

$$-\frac{8}{9} < x < -\frac{2}{9}$$

$$Df : \left(-\frac{8}{9}, -\frac{2}{9} \right) \text{ (A)}$$

Q2:

$$\lim_{x \rightarrow -4} \frac{x^2 - 6x + 8}{\sqrt{x^2 - 8x + 16}}$$

sol :-

$$\lim_{x \rightarrow -4} \frac{x^2 - 6x + 8}{|x - 4|} \implies \text{نلاحظ ان } (\sqrt{x})^2 = x, \text{ but } \sqrt{x^2} = |x|$$

$$|x - 4| \begin{matrix} \leftarrow -(x-4) & (x-4) \rightarrow \end{matrix} \implies |x - 4| \text{ تعريف}$$

$$\text{بما ان } (x \rightarrow -4) \text{ نختار } -(x-4)$$

$$\lim_{x \rightarrow -4} \frac{x^2 - 6x + 8}{-(x-4)}$$

Don't miss the change..

Be part of it !!



$$\lim_{x \rightarrow 4} \frac{(x-4)(x-2)}{-(x-4)} \quad \text{تأجيل البسط}$$

$$\frac{4-2}{-1} = -2 \quad \textcircled{C}$$

Q3 :-

$$f(x) = \pi/2 + \cot^{-1}(x-2)$$

SOL :-

$$f(2) = \pi/2 + \cot^{-1}(0) \quad \text{تعويض مباشر}$$

$$= \pi/2 + \pi/2 = \pi \quad \textcircled{C}$$

Q4 :-

$$f(x) = 4[x]$$

SOL :-

* يجب أن تكون جواب $[f]$ عدد صحيح وبالكافي
تكون طريقة الحل بالتجريب :-

$$f(1) = 4, \quad f(-2) = -8$$

$$f(0) = 0, \quad f(-1) = -4$$

$$Rf \{0, \pm 4, \pm 8, \pm 12, \dots\}$$



Q5:-

$$f(x) = 6\sin^{-1}x^2$$

: Absolute *

Sol:-

$$0 \leq \sin^{-1}x^2 \leq \pi/2$$

نطاق الدالة العكسية للافتراض

$$0 \leq 6\sin^{-1}x^2 \leq 3\pi$$

Range $\sin^{-1}x$

$$-\pi/2 \leq x \leq \pi/2$$

Range $\sin^{-1}x^2$

$$Rf : [0, 3\pi] \text{ (A)}$$

$$0 \leq x \leq \pi/2$$

Q6:-

$$\lim_{x \rightarrow 3^+} \left(\frac{1}{x-3} - \frac{27}{x^2-27} \right)$$

Sol:-

نطاق الدالة العكسية للافتراض * $\frac{1}{x-3}$

$$\lim_{x \rightarrow 3^+} \left(\frac{x^2+3x+9}{x^2-27} - \frac{27}{x^2-27} \right)$$

$$\lim_{x \rightarrow 3^+} \left(\frac{x^2+3x-9}{x^2-27} - \frac{27}{x^2-27} \right)$$

$$\lim_{x \rightarrow 3^+} \frac{(x-3)(x+6)}{(x-3)(x^2+3x+9)} = \frac{1}{3} \text{ (A)}$$



Q7:-

$$2\sin^{-1}\left(\frac{\sqrt{7}}{19}\right) + 2\cos^{-1}\left(\frac{\sqrt{7}}{19}\right)$$

sol:- (مجموعه 2 جیب)

$$2\left(\sin^{-1}\frac{\sqrt{7}}{19} + \cos^{-1}\frac{\sqrt{7}}{19}\right)$$

$$2 * \pi / 2 = \pi \quad \textcircled{A}$$

Q8:-

$$Df = (4, 7], g(x) = f(2-x)$$

sol:-

$$4 < x \leq 7$$

① متغیر المبدأ 4 ی -1 و بالایی

تقلیل المبدأ 4

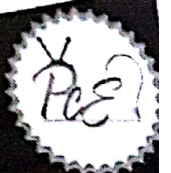
$$-4 > x \geq -7$$

② مجموع 2 المبدأ 4

$$-2 > \underbrace{x+2}_{g(x)} \geq -5$$

 $g(x)$

$$Dg = [-5, -2] \quad \textcircled{B}$$



Q9 :-

$$\lim_{x \rightarrow \infty} \frac{\sqrt{25x^2 + x + 1} + 14x + 8}{6x + 4}$$

Sol :-

$$\lim_{x \rightarrow -\infty} \frac{5|x| + 14x}{6x} \rightsquigarrow |x| \text{ يجب إعادة تعريفه}$$

$$\begin{array}{c} -x \quad x \\ \leftarrow \quad - \quad + \quad \rightarrow \quad |x| \\ 0 \end{array}$$

نأخذ $(-x)$ لأن $(x \rightarrow -\infty)$

$$\lim_{-x \rightarrow -\infty} \frac{-5x + 14x}{6x}$$

$$\lim_{x \rightarrow -\infty} \frac{9x}{6x} = \frac{3}{2} \quad \textcircled{A}$$

Q10 :-

$$y=0, x=2, f(x) = \frac{ax^3 - x(3x+1)}{x^3 - b}$$

Sol :-

$$\lim_{x \rightarrow \pm\infty} \frac{ax^3 - x(3x+1)}{x^3 - b}$$

$$\lim_{x \rightarrow \pm\infty} \frac{ax^3}{x^3} = a$$

$$\therefore a = y = 0 \rightarrow a = 0$$

نجد (b) عن طريق أمثفار المقام، لأن أمثفار المقام تجعل النهاية تساوي $\pm\infty$



$$x^3 - b = 0$$

$$8 - b = 0 \rightarrow b = 8$$

$$a + b = 8 \quad \textcircled{B}$$

Q11:-

$$f(x) = x^2, \quad g(x) = \begin{cases} 2x, & x \leq 4 \\ x+1, & x > 4 \end{cases}$$

Sol:-

$$(f \circ g)(x) = f(g(x)) = \begin{cases} 4x^2, & x \leq 4 \\ (x+1)^2, & x > 4 \end{cases} \quad \textcircled{B}$$

Q12 :-

$$f(x) = \cos^{-1}(5x+1)$$

Sol:-

$$Df = D\cos^{-1} = [-1, 1]$$

$$-1 \leq 5x+1 \leq 1$$

$$(1) \text{ } \cos^{-1} \text{ (مجال الأعداد)}$$

$$-2 \leq 5x \leq 0$$

$$(2) \text{ } \cos^{-1} (5x+1) \text{ (مجال الأعداد)}$$

$$-\frac{2}{5} \leq x \leq 0$$

$$Df = \left[-\frac{2}{5}, 0 \right] \quad \textcircled{A}$$

Don't miss the change..
Be part of it !!



Q14:-

قوانين جيب دلتا :-

$$f(x) = \csc^{-1} x$$

$$* \sec^{-1} \theta = \cos^{-1} \frac{1}{\theta}$$

sol :-

$$* \csc^{-1} \theta = \sin^{-1} \frac{1}{\theta}$$

$$f(3) = \csc^{-1} 3$$

$$f(3) = \sin^{-1} \frac{1}{3} \quad \textcircled{D}$$

Q13:-

$$f(x) = x^3 \ln |x|$$

sol:-

$$f(-x) = -x^3 \ln |-x|$$

$$= -x^3 \ln |x|$$

$$-f(-x) = -x^3 \ln |x|$$

$$f(-x) = -f(x) \implies \text{odd function}$$

$\therefore f$ is odd $\textcircled{C}$

Q15:-

$$f(x) = \frac{1}{\sqrt{9-x^2}}$$

sol:-

$$x^2 \geq 0 \quad \text{التعبير دلتا}$$

$$\text{كلها}$$



نختوب المتباينة بـ 1- وبالتالي 1)

$$-x^2 \leq 0$$

نختوب 9 الى طرفي المتباينة 2)

$$9 - x^2 \leq 9$$

نأخذ الجذر التربيعي للمتباينة 3)

$$\sqrt{9-x^2} \leq \sqrt{9}$$

$$\sqrt{9-x^2} \leq 3$$

نقلب $\sqrt{9-x^2}$ و 3 وبالتالي نقاب 4)

المتباينة $\therefore$

$$\frac{1}{\sqrt{9-x^2}} \geq \frac{1}{3}$$

$$Rf = \left[\frac{1}{3}, \infty \right) \quad \textcircled{D}$$

Q16 :-

$$f(x) = x^2 + 4x + 7, \quad x \in (-\infty, K]$$

Sol :-

الاقتبان هو اقتبان تربيعي لذلك :-

$$Df = (-\infty, K]$$

من K نأخذ الرأس وبالتالي نجد K عن طريق

$$\left(\frac{-b}{2a} \right)$$



$$\therefore \left(\frac{-b}{2a} \right) = \frac{-4}{2} = -2 \quad \textcircled{B}$$

Q17:-

$$f(x) = \begin{cases} 3-x, & x < 0 \\ 2+e^{-x}, & x \geq 0 \end{cases}$$

SOL:-

یہ دو پٹاں ملے گی اور ان کے انورس فائنڈ کرنا ہوں گے۔

$f_1 = 3-x \rightarrow y_1 = 3-x$ $x = 3-y_1$ $y_1 = 3-x$	$f_2 = 2+e^{-x} \rightarrow y_2 = 2+e^{-x}$ $x = 2+e^{-x}$ $e^{-y_2} = x-2$ $-y_2 = \ln(x-2) \rightarrow y_2 = -\ln(x-2)$
---	--

$$f^{-1}(x) = \begin{cases} 3-x, & x > 3 \\ -\ln(x-2), & 2 < x \leq 3 \end{cases} \quad \textcircled{A}$$



Q18 :-

$$e^{7x} > 2e^{3x}$$

sol :-

$$\ln e^{7x} > \ln 2e^{3x} \quad (\ln x^n = n \ln x)$$

$$7x \ln e > \ln 2 + \ln e^{3x} \quad (\ln ab = \ln a + \ln b)$$

$$7x > \ln 2 + 3 \ln e^x$$

$$7x > \ln 2 + 3x$$

نقل المتباينة

$$4x > \ln 2$$

$$x > \frac{\ln 2}{4}$$

$$x > 1/4 \ln 2$$

(نستخرج الناحية السالبة بالرفعة مربعة)

$$x > \ln 2^{1/4}$$

$$x > \ln^4 \sqrt{2}$$

$$\therefore (\ln^4 \sqrt{2}, \infty) \quad \textcircled{D}$$



Q 19 :-

$$f(x) = \frac{1}{x-2}, g(x) = \frac{1}{x}$$

Sol :-

لايجاد (D) اي اقليم مرتب، يجب اتباع القاعدة التالية :-

$$D_{f \circ g} = D_g \cap D_h$$

$$h = f \circ g$$

$$h = \frac{1}{\frac{1}{x} - 2}$$

$$h = \frac{x}{1-2x} \quad (\text{اقليم مسموح})$$

$$D_h = R / \{x=1/2\}$$

$$-D_h = R / x=1/2$$

$$g = \frac{1}{x} \quad (\text{اقليم مسموح})$$

$$D_g = R / x=0$$

$$\therefore D_g \cap D_h$$

$$= R / \{0, 1/2\} \quad (C)$$



Q 20 :-

$$f(x) = -4 + \ln(2x+3)$$

Sol:-

$$x\text{-intercept} \rightarrow y=0$$

अनुमान

$$y=0 \quad (f(x)=0)$$

$$-4 + \ln(2x+3) = 0$$

$$\ln(2x+3) = 4$$

$$e^4 = 2x+3$$

$$2x = e^4 - 3$$

$$x = \frac{e^4 - 3}{2} \quad \textcircled{B}$$



1. The domain of the function $f(x) = \ln(3 - |9x + 5|)$ is

a. ☒ A. $(-\frac{8}{9}, -\frac{2}{9})$

B. $\mathbb{R} \setminus (-\frac{8}{9}, -\frac{2}{9})$

c. ☐ C. $\mathbb{R} \setminus [-\frac{8}{9}, -\frac{2}{9}]$

d. ☒ D. $[-\frac{8}{9}, -\frac{2}{9}]$

2. $\lim_{x \rightarrow 4^-} \frac{x^2 - 6x + 8}{\sqrt{x^2 - 8x + 16}} =$

A. does not exist

B. 0

☒ C. -2

D. 22

3. Given that $f(x) = \frac{\pi}{2} + \cot^{-1}(x-2)$ then $f(2) =$

A. undefined

B. 2π

☒ C. π

D. $\frac{5}{2}\pi$

4. (The greatest integer less than or equals x) The range of $f(x) = 4[x]$ is

☒ A. $\{0, \pm 4, \pm 8, \pm 12, \dots\}$

C. $(-\infty, \infty)$

B. $\mathbb{R} \setminus \{0, \pm 1, \pm 2, \pm 3, \dots\}$

D. $\{0, \pm 1, \pm 2, \pm 3, \dots\}$

5. The range of the function $f(x) = 6 \sin^{-1}(x^2)$ is

☒ A. $[0, 3\pi]$

B. $[-3\pi, 0]$

☒ C. $[-3\pi, 3\pi]$

D. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

6. $\lim_{x \rightarrow 3^+} \left(\frac{1}{x-3} - \frac{27}{x^3-27} \right) =$

☒ A. $\frac{1}{3}$

B. -1

C. $-\frac{1}{3}$

D. 1

7. The value of $2 \sin^{-1}\left(\frac{\sqrt{7}}{19}\right) + 2 \cos^{-1}\left(\frac{\sqrt{7}}{19}\right) =$

A. $\frac{1}{4}\pi$

B. 2π

☒ C. π

D. $\frac{1}{2}\pi$

8. If the domain of the function $y = f(x)$ is $(4, 7]$ then the domain of $g(x) = f(2-x)$ is

A. $[4, 7)$

☒ B. $[-5, -2)$

C. $(-5, -2]$

D. $(4, 7]$

9. $\lim_{x \rightarrow -\infty} \left(\frac{\sqrt{25x^2 + x + 1} + 14x + 8}{6x + 4} \right) =$

A. $\frac{3}{2}$

B. ∞

C. $-\infty$

D. $-\frac{5}{6}$

10. If $y = 0$ and $x = 2$ are horizontal and vertical asymptotes (respectively) of the function

$$f(x) = \frac{ax^3 - x(3x + 1)}{x^3 - b} \quad a + b =$$

A. 3

B. 8

C. 5

D. 11

11. Given that $f(x) = x^2$ and $g(x) = \begin{cases} 2x & , x \leq 4 \\ x+1 & , x > 4 \end{cases}$ then $(f \circ g)(x) =$

A. $\begin{cases} 4x^2 & , x < 4 \\ (x+1)^2 & , x \geq 4 \end{cases}$

C. $\begin{cases} 4x^2 & , x \leq 16 \\ (x+1)^2 & , x > 16 \end{cases}$

B. $\begin{cases} 4x^2 & , x \leq 4 \\ (x+1)^2 & , x > 4 \end{cases}$

D. $\begin{cases} 4x^2 & , x < 16 \\ (x+1)^2 & , x \geq 16 \end{cases}$

12. The domain of $f(x) = \cos^{-1}(5x + 1)$ is

A. $[-\frac{2}{5}, 0]$

B. $[-\frac{1}{5}, 0]$

C. $[-1, 1]$

D. $[0, \frac{2}{5}]$

13. The function $f(x) = x^3 \ln|x|$ is.

A. neither odd nor even

B. even

C. odd

D. negative for all $x \in \text{Domain}(f)$

14. Given that $f(x) = \csc^{-1} x$ then $f(3) =$

A. $\frac{1}{\cos 3}$

B. $\frac{1}{\sin 3}$

C. $\cos^{-1}\left(\frac{1}{3}\right)$

D. $\sin^{-1}\left(\frac{1}{3}\right)$

15. The range of the function $f(x) = \frac{1}{\sqrt{9-x^2}}$ is

A. $(\frac{1}{3}, \infty)$

B. $(0, 3)$

C. $[0, 3]$

D. $[\frac{1}{3}, \infty)$

16. If the $f(x) = x^2 + 4x + 7$, $x \in (-\infty, k]$ is one to one function, then the largest possible value of k is

A. 2

B. -2

C. -4

D. 4

17. If $f(x) = \begin{cases} 3-x, & x < 0 \\ 2+e^{-x}, & x \geq 0 \end{cases}$ then $f^{-1}(x) =$

A. $\begin{cases} 3-x, & x > 3 \\ -\ln(x-2), & 2 < x \leq 3 \end{cases}$

B. $\begin{cases} 3-x, & x > 3 \\ -2+\ln x, & 2 < x \leq 3 \end{cases}$

C. $\begin{cases} x-3, & x > 3 \\ -\ln(x-2), & 2 < x \leq 3 \end{cases}$

D. $\begin{cases} x-3, & x > 3 \\ -2+\ln x, & 2 < x \leq 3 \end{cases}$

18. The solution set of the inequality $e^{7x} > 2e^{3x}$ is

A. $(-\infty, \ln^4 \sqrt{2})$ B. $(0, \frac{1}{4} \ln 2)$ C. $(-\frac{1}{4} \ln 2, 0)$ D. $(\ln^4 \sqrt{2}, \infty)$

19. Given that $f(x) = \frac{1}{x-2}$ and $g(x) = \frac{1}{x}$ then the domain of the function $f \circ g$ is

A. $\mathbb{R} \setminus \{0, 2\}$ B. $\mathbb{R} \setminus \{\frac{1}{2}\}$ C. $\mathbb{R} \setminus \{0, \frac{1}{2}\}$ D. $\mathbb{R} \setminus \{0\}$

20. The x -intercept of the function $f(x) = -4 + \ln(2x+3)$ is

A. $\frac{1}{2}e^4$ B. $\frac{1}{2}(e^4 - 3)$ C. $\frac{1}{2}(3 - e^4)$ D. 0

$\frac{1}{2} \ln 5 = \ln \sqrt{5}$

$$Q_1: f(x) = \ln(3 - |9x + 5|)$$

Sol:-

$$3 - |9x + 5| > 0 \rightarrow \text{Domain } \ln f = f > 0$$

$$|9x + 5| < 3$$

$$-3 < 9x + 5 < 3 \rightarrow \text{خواص القيمة المطلقة}$$

$$-8 < 9x < -2$$

$$-\frac{8}{9} < x < -\frac{2}{9}$$

$$Df: \left(-\frac{8}{9}, -\frac{2}{9}\right) \text{ (A)}$$

$$Q_2: \lim_{x \rightarrow 4} \frac{x^2 - 6x + 8}{\sqrt{x^2 - 8x + 16}}$$

Sol:-

$$\lim_{x \rightarrow 4} \frac{x^2 - 6x + 8}{\sqrt{(x-4)^2}} \rightarrow \text{تحليل ما أسفل الجذر}$$

$$\lim_{x \rightarrow 4} \frac{x^2 - 6x + 8}{|x-4|} \rightarrow (\sqrt{x})^2 = x, \text{ but } \sqrt{x^2} = |x|$$

$$|x-4| \xrightarrow[-(x-4)]{(x-4)} \rightarrow |x-4| \text{ مادة تعريف}$$

$$\text{بما أنه } x \rightarrow 4^- \text{ فختار } -(x-4)$$

$$\lim_{x \rightarrow 4} \frac{x^2 - 6x + 8}{-(x-4)}$$

$$\lim_{x \rightarrow 4} \frac{(x-4)(x-2)}{-(x-4)} \rightarrow \text{تحليل البسط}$$

$$\frac{4-2}{-1} = -2 \text{ (C)}$$



$$Q_3: f(x) = \frac{\pi}{2} + \cot^{-1}(x-2)$$

Sol:-

$$f(2) = \frac{\pi}{2} + \cot^{-1}(0) \rightarrow \text{تعويض مباشر}$$

$$= \frac{\pi}{2} + \frac{\pi}{2} = \pi \quad (C)$$

$$Q_4: f(x) = 4[x]$$

Sol:-

يجب أن يكون جواب $[f]$ عدد صحيح

وبالتالي تكون طريقة الحل بالتجريب:

$$f(1) = 4, \quad f(-2) = -8$$

$$f(0) = 0, \quad f(-1) = -4$$

$$\therefore R_f \in \{0, \pm 4, \pm 8, \pm 12, \dots\} \quad (A)$$



Q5: $f(x) = 6 \sin^{-1} x^2$

Sol:-

المجال المطابق :-

$$0 \leq \sin^{-1} x^2 \leq \frac{\pi}{2}$$

Range $\sin^{-1} x$:

لجعلها مطابقة للصيغة الافتراضية $0 \leq x \leq 1$

$$-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

$$0 \leq \sin^{-1} x^2 \leq \frac{\pi}{2}$$

نكتب

$$0 \leq 6 \sin^{-1} x^2 \leq 3\pi$$

$f \swarrow$

Range $\sin^{-1} x^2$

$$0 \leq x \leq \frac{\pi}{2}$$

$$R_f: [0, 3\pi] \text{ (A)}$$

الترتيب x يلفس الفتحة التالية

Q6: $\lim_{x \rightarrow 3^+} \left(\frac{1}{x-3} - \frac{27}{x^3-27} \right)$

Sol:-

توحيد المقامات - كما نرى ضرب $\frac{1}{x-3}$ بالمقام (x^3-27) بالبرهان التالي

$$\lim_{x \rightarrow 3^+} \left(\frac{x^2+3x+9}{x^3-27} - \frac{27}{x^3-27} \right)$$

$$\lim_{x \rightarrow 3^+} \frac{x^2+3x-18}{x^3-27}$$

* قوانين المرافقة التاليفي:

$$\lim_{x \rightarrow 3^+} \frac{(x-3)(x+6)}{(x-3)(x^2+3x+9)}$$

تحليل البسط والمقام $\rightarrow$

$$(x-\alpha) \xrightarrow{\text{مرافقة}} x^2 + \alpha x + \alpha^2$$

$$= \frac{9}{27} = \frac{1}{3} \text{ (A)}$$

$$(x-\alpha)(x^2+\alpha x+\alpha^2) \xrightarrow{\text{تبسيط}} x^3 - \alpha^3$$

$$(x+\alpha) \xrightarrow{\text{مرافقة}} x^2 - \alpha x + \alpha^2$$

$$(x+\alpha)(x^2-\alpha x+\alpha^2) \xrightarrow{\text{تبسيط}} x^3 + \alpha^3$$



$$Q7: 2 \sin^{-1}\left(\frac{\sqrt{7}}{4}\right) + 2 \cos^{-1}\left(\frac{\sqrt{7}}{4}\right)$$

sol:-

نأخذ 2 كامل مشترك

$$2 \left(\sin^{-1}\left(\frac{\sqrt{7}}{4}\right) + \cos^{-1}\left(\frac{\sqrt{7}}{4}\right) \right)$$

$$\boxed{\sin^{-1}\theta + \cos^{-1}\theta = \frac{\pi}{2}}$$

قانون يجب حفظ ←

$$\therefore 2 * \frac{\pi}{2} = \pi \quad \textcircled{C}$$

$$Q8: Df = (4, 7] , g(x) = f(2-x)$$

sol:-

$$4 < x \leq 7$$

① نظرب المتباينة بر 1 - وبالتالي تقلبت

إشارات المتباينة.

$$-4 > -x \geq -7$$

② نضيف 2 للمتباينة

$$-2 > 2-x \geq -5$$

$g(x)$

$$\therefore Dg = [-5, -2) \quad \textcircled{B}$$

يجب الوصول

إلى صيغة $2-x$ ②



Q9: $\lim_{x \rightarrow -\infty} \frac{\sqrt{25x^2 + x + 1} + 14x + 8}{6x + 4}$

Sol: -

$\lim_{x \rightarrow -\infty} \frac{5|x| + 14x}{6x} \rightarrow |x|$ - ليجب، المادة تعرف $|x|$

$\xleftarrow{-x} \quad \xrightarrow{+} \quad |x|$

$x \rightarrow -\infty$ نأخذ $-x$

$\lim_{x \rightarrow -\infty} \frac{-5x + 14x}{6x}$

$\lim_{x \rightarrow -\infty} \frac{9x}{6x} = \frac{3}{2} \quad \textcircled{A}$

Q10: $y=0, x=2, f(x) = \frac{ax^3 - x(3x+1)}{x^3 - b}$

Sol: -

$\lim_{x \rightarrow \pm\infty} \frac{ax^3 - x(3x+1)}{x^3 - b}$

$\lim_{x \rightarrow \pm\infty} \frac{ax}{x^3} = a$

$\therefore a = y = 0 \Rightarrow \boxed{a = 0}$

نجد b كطريقة أصفار المقام

لأن أصفار المقام تجعل النسبة تساوي $\pm\infty$

$x^3 - b = 0$

$8 - b = 0 \Rightarrow \boxed{b = 8}$

$\therefore a + b = 8 \quad \textcircled{B}$



$$Q_{11}: f(x) = x^2, \quad g(x) = \begin{cases} 2x & , x \leq 4 \\ x+1 & , x > 4 \end{cases}$$

Sol:-

$$(f \circ g)(x) = f(g(x)) = \begin{cases} 4x^2 & , x \leq 4 \\ (x+1)^2 & , x > 4 \end{cases} \quad (B)$$

$$Q_{12}: f(x) = \cos^{-1}(5x+1)$$

Sol:-

$$D_f = D_{\cos^{-1}} = [-1, 1]$$

$$-1 \leq 5x+1 \leq 1$$

① نظر 1 من جميع الأضلاع

$$-2 \leq 5x \leq 0$$

② نقب 5 (جميع الأضلاع)

$$-\frac{2}{5} \leq x \leq 0$$

$$D_f = \left[-\frac{2}{5}, 0\right] \quad (A)$$

نحسب الوصول إلى

x لوحدتها: ① ②



Q13: $f(x) = x^3 \ln|x|$

Sol: -

$$f(-x) = -x^3 \ln|-x|$$

$$= -x^3 \ln|x|$$

$$-f(x) = -x^3 \ln|x|$$

$$f(-x) = -f(x) \rightarrow \text{تحقق شرط الـ odd}$$

$\therefore f$ is odd ©

Q14: $f(x) = \csc^{-1} x$

Sol: -

$$f(3) = \csc^{-1} 3$$

قوانين يجب حفظها!

$$* \sec^{-1} \theta = \cos^{-1} \frac{1}{\theta}$$

$$* \csc^{-1} \theta = \sin^{-1} \frac{1}{\theta}$$

$$\therefore f(3) = \sin^{-1} \frac{1}{3} \quad \text{© D}$$



$$Q_{15}: f(x) = \frac{1}{\sqrt{9-x^2}}$$

So:-

$$x^2 \geq 0 \rightarrow \text{التربيع يغير السالب}$$

بحسب الوصول إلى الصيغة $\frac{1}{\sqrt{9-x^2}}$
منه طريقه الخطوات ① ② ③ ④

① نضرب المتباينة بـ -1 - وبالنسبة لتقلب الإشارة.

$$-x^2 \leq 0$$

② نضيف 9 إلى طرفي المتباينة.

$$9 - x^2 \leq 9$$

③ نأخذ الجذر التربيعي لطرفي المتباينة.

$$\sqrt{9-x^2} \leq \sqrt{9}$$

$$\sqrt{9-x^2} \leq 3$$

④ نقلب $\sqrt{9-x^2}$ و 3 (طرفي المتباينة)

وبالتالي تتقلب الإشارة للمتباينة.

$$\frac{1}{\sqrt{9-x^2}} \geq \frac{1}{3}$$

$$\therefore R_f = \left[\frac{1}{3}, \infty\right) \quad D$$



Q16: $f(x) = x^2 + 4x + 7, x \in (-\infty, k]$

sol:-

الافتراض هو افتراضه شريطة لانه:

$Df = (-\infty, k]$

حيث k تمثل الرأس وبالتالي نجد k منه طريقه قانونه

رأس القطر وهو $(\frac{-b}{2a})$

$\therefore (\frac{-b}{2a}) = \frac{-4}{2} = -2$ (B)

Q17: $f(x) = \begin{cases} 3-x & , x < 0 \\ 2+e^{-x} & , x \geq 0 \end{cases}$

sol:-

خطوات إيجاد f^{-1} هي: نجعل x مكان y ونجعل x ثم نجعل y موضوعاً للقانون.

$f_1 = 3-x \Rightarrow y_1 = 3-x$

$x = 3-y_1$

$y_1 = 3-x$

$f_2 = 2+e^{-x} \Rightarrow y_2 = 2+e^{-x}$

$x = 2+e^{-x}$

$e^{-y_2} = x-2$

$-y_2 = \ln(x-2) \Rightarrow y_2 = -\ln(x-2)$

$\therefore f^{-1}(x) = \begin{cases} 3-x & , x > 3 \\ -\ln(x-2) & , 2 < x \leq 3 \end{cases}$ (A)

* ملاحظة: Df مع وجودها الامكان

وبالتالي لا داعي للايجاد



$$Q. 18: e^{7x} > 2e^{3x}$$

Sol:-

$$\ln e^{7x} > \ln 2e^{3x}$$

$$\boxed{\ln x^n = n \ln x}$$

نستخدم خواص اللوغاريتمية $\Leftrightarrow$

$$7x \ln e > \ln 2 + \ln e^{3x} \rightarrow \boxed{\ln ab = \ln a + \ln b} \leftarrow \text{خاصية أخرى}$$

$$7x > \ln 2 + 3x \ln e$$

$$7x > \ln 2 + 3x$$

نحل المتباينة

$$4x > \ln 2$$

$$x > \frac{\ln 2}{4}$$

$$x > \frac{1}{4} \ln 2$$

نستخدم الخاصية السابقة بطريقة معاكسة

$$x > \ln 2^{\frac{1}{4}}$$

$$x > \ln^4 \sqrt{2}$$

$$\therefore (\ln^4 \sqrt{2}, \infty) \text{ (D)}$$



$$Q19: f(x) = \frac{1}{x-2}, \quad g(x) = \frac{1}{x}$$

Sol: -

للايجاد D أي $f \circ g$ مركبة بحسب القاعدة التالية:

$$D_{f \circ g} = D_g \cap D_h$$

$$h = f \circ g \quad \text{حيث:}$$

$$h = f \circ g(x) = f(g(x))$$

$$h = \frac{1}{\frac{1}{x} - 2} \rightarrow \text{نوجد المقامات}$$

$$h = \frac{1}{\frac{1-2x}{x}}$$

$$h = \frac{x}{1-2x}$$

$$h \rightarrow \text{مقرّر}$$

$$D_h = \mathbb{R} \setminus \text{أصفار المقام}$$

$$= \mathbb{R} \setminus x = \frac{1}{2}$$

$$D_g \text{ نجد}$$

$$g = \frac{1}{x} \rightarrow \text{مقرّر منقسم}$$

$$D_g = \mathbb{R} \setminus x = 0$$

$$\therefore D_g \cap D_h$$

$$= \mathbb{R} \setminus \{0, \frac{1}{2}\} \quad \textcircled{C}$$



$$Q_{20}: f(x) = -4 + \ln(2x+3)$$

Sol:-

$$x\text{-intercept} \rightarrow y = 0$$

المقطع y اليساري

$$y = 0 \Rightarrow f(x) = 0$$

$$f(x) = 0 = -4 + \ln(2x+3)$$

$$4 = \ln(2x+3)$$

$$e^4 = 2x+3$$

$$2x = e^4 - 3$$

$$x = \frac{e^4 - 3}{2}$$

$$x = \frac{1}{2}(e^4 - 3) \text{ (B)}$$



Solution of Mid Exam - Summer Sem. 2016/2017:

Q-1:- Evaluate the Limit:

$$\lim_{x \rightarrow 4} \frac{16}{x^2 - 16}$$

$$* \lim_{x \rightarrow 4} \frac{16}{x^2 - 16} = \frac{16}{0} \quad \left/ \quad \begin{array}{l} x^2 - 16 = 0 \\ x = +4 \end{array} \right.$$

$$\lim_{x \rightarrow 4^+} \frac{16}{x^2 - 16} = +\infty$$

$$\lim_{x \rightarrow 4^-} \frac{16}{x^2 - 16} = -\infty$$

$$\Rightarrow \lim_{x \rightarrow 4} \text{D.N.E}$$

Q-2: Evaluate the Limit:

$$\lim_{x \rightarrow \infty} \sin^{-1} \left(\frac{x+1}{1+2x} \right)$$

$$* \sin^{-1} \left(\lim_{x \rightarrow \infty} \left(\frac{x+1}{1+2x} \right) \right) \Rightarrow \lim_{x \rightarrow \infty} \frac{x}{2x} = \frac{1}{2}$$

$$\Rightarrow \sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{6}$$

Q-3: Find the value(s) of x (if any) that satisfy $\ln(x+1) = e$

$$* \ln(x+1) = e$$

$$x+1 = e^e$$

$$x = e^e - 1$$

Q-4-: Evaluate the Limit :

$$\lim_{x \rightarrow 0} \frac{\tan(3x) + 3x}{\sin(2x)}$$

$$\star \lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(2x)} + \lim_{x \rightarrow 0} \frac{3x}{\sin(2x)}$$

$$\Rightarrow \frac{3}{2} + \frac{3}{2} = \frac{6}{2} = 3$$

Q-5- Evaluate the Limit :

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x} \times \frac{\sqrt{x+1} + 1}{\sqrt{x+1} + 1}$$

$$\lim_{x \rightarrow 0} \frac{x+1-1}{x(2)} \Rightarrow \lim_{x \rightarrow 0} \frac{x}{2x} = \lim_{x \rightarrow 0} \frac{1}{2} = \frac{1}{2}$$

Q-6-: Evaluate the Limit:

$$\lim_{x \rightarrow 1} \frac{x \sin(2x-2) - x^2}{2x^2}$$

$$\lim_{x \rightarrow 1} \frac{x \sin(2x-2)}{2x^2} - \lim_{x \rightarrow 1} \frac{x^2}{2x^2} \Rightarrow \lim_{x \rightarrow 1} \frac{\sin(2x-2)}{2x} - \lim_{x \rightarrow 1} \frac{1}{2}$$

$$\lim_{x \rightarrow 1} \frac{\sin(2x-2)}{2x-2} \cdot \frac{2x-2}{2x} - \frac{1}{2} \Rightarrow \lim_{x \rightarrow 1} \frac{\sin(2x-2)}{2x-2} \cdot \frac{(x-1)}{x} - \frac{1}{2}$$

$$\frac{2x}{2x-2} \quad (1 \times 0) - \frac{1}{2} = -\frac{1}{2}$$

Don't miss the change.
Be part of it !!

Q-7- : Evaluate the Limit :

$$\lim_{x \rightarrow -\infty} \sqrt{x^2 - 2} + x$$

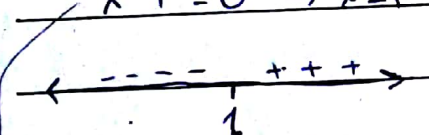
$$\lim_{x \rightarrow -\infty} \sqrt{x^2 - 2} + x = \infty - \infty$$

$$\lim_{x \rightarrow -\infty} \left(\frac{\sqrt{x^2 - 2} + x}{1} \right) \times \left(\frac{\sqrt{x^2 - 2} - x}{\sqrt{x^2 - 2} - x} \right)$$

$$\lim_{x \rightarrow -\infty} \frac{x^2 - 2 - x^2}{\sqrt{x^2 - 2} - x} \rightarrow \lim_{x \rightarrow -\infty} \frac{-2}{\sqrt{x^2 - 2} - x} = 0$$

Q-8- : Study the continuity of the function

$$f(x) = \frac{\ln(x-1)}{x-2}$$

$x-1=0 \rightarrow x=1$ 	$x-2$ $\text{Domain} = (-\infty, \infty)$ $\text{Cont. on} \Rightarrow (1, \infty) \cup (-\infty, \infty)$ the function is continuity $(1, \infty)$
--	--

Q-9- : Find the vertical asymptote(s) (if any) for the function :

$$f(x) = \csc(2x)$$

Sol. Q-9- : $f(x) = \csc(2x)$

$$f(x) = \frac{1}{\sin(2x)}$$

$$\sin(2x) = 0$$

$$x = 0, \frac{\pi}{2}, \frac{3\pi}{2}, \dots, \frac{n+1}{2}\pi$$

All values of x is vertical asymptotes.

Q-10- : Find the horizontal asymptote(s)

(if any) for the function:

$$f(x) = xe^x$$

$$\lim_{x \rightarrow \infty} xe^x \Rightarrow \infty \cdot \infty = +\infty$$

$$\lim_{x \rightarrow -\infty} xe^x \Rightarrow -\infty \cdot 0 = \lim_{x \rightarrow -\infty} \frac{e^x}{\frac{1}{x}} = \frac{0}{0}$$

Horizontal asymptotes isn't found.



Al Balqa Applied University
Faculty of Engineering Technology
Department Physics and Basic science
Midterm Exam
Calculus 1

KEY

Student Name:

Lecture time:

Student Number:

Instructor:

Section 1: Multiple choice questions. (30 marks)
Write out the correct answer symbol in the table. (Use capital letters)

Question Number	1	2	3	4	5	6
Answer symbol	C	D	B	B	A	A
Question Number	7	8	9	10	11	12
Answer symbol	A	C	D	D	B	C

Q1. If $f(x) = 3x - 2$ and $(f \circ g)(x) = 12x^2 - 6x + 16$, then $g(0) =$

- A) 2 B) 5 C) 6 D) 7

Q2. The domain for the function $f(x) = \sqrt{9 - x^2} \sec x$, is:

- A) $[-3, 3] - \left\{\frac{\pi}{2}\right\}$ B) $[-3, 3] - \{0\}$ C) $[-3, 3]$ D) $[-3, 3] - \left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\}$

Q3. Let $f(x) = 2x^2 + 5x + 3$. Then the value of x such that $f^{-1}(x) = 0$, is

- A) 0 B) 3 C) 2 D) 1

Q4. $\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2}}{x^2} =$

- A) $-\frac{1}{2}$ B) $\frac{1}{2}$ C) $\frac{3}{2}$ D) 0

Q5. $\lim_{x \rightarrow \infty} \frac{\sin^2 x + 3x^2}{6x^2 + \cos^2 x} =$

- A) $\frac{1}{2}$ B) $-\frac{1}{2}$ C) does not exist D) ∞

Q6. The function $f(x) = \sqrt[3]{x^3 - x}$ is

- A) Odd B) Even C) Neither even nor odd D) Symmetric about y axis

Q7. If $\lim_{x \rightarrow 0} f(x) = 1$, $\lim_{x \rightarrow 0} g(x) = -1$, then $\lim_{x \rightarrow 0} \frac{2f(x) - g(x)}{(f(x) + 7)^{\frac{1}{3}}} =$

- A) $\frac{3}{4}$ B) $-\frac{3}{4}$ C) $\frac{3}{2}$ D) $-\frac{3}{2}$



K E

Q8. The horizontal asymptote(s) (if any) for the function $f(x) = \frac{\sqrt{x^2+1}-1}{x-1}$, is (are):

- A) $y = -1$ B) $y = 1$ C) $y = -1, y = 1$ D) No horizontal asymptotes

Q9. The vertical asymptote(s) (if any) for the function $f(x) = \frac{(x^2-1)^3}{(x^2-1)^2}$, is (are):

- A) $x = -1, x = 1$ B) $x = 1$ C) No vertical asymptotes D) $x = -1$

Q10. The range for the function, $f(x) = \sqrt{\frac{x-1}{x+2}}$ is:

- A) $\mathbb{R} - \{1\}$ B) $\mathbb{R} - \{-1\}$ C) $\mathbb{R} - \{-2\}$ D) $\mathbb{R} - \{1\}$

Q11. The value of k , that makes $\lim_{x \rightarrow 1} \frac{\sqrt{2x^2-kx-3}}{x-1}$, exists and finite is:

- A) $k = 2$ B) $k = 3$ C) $k = 1$ D) $k = 4$

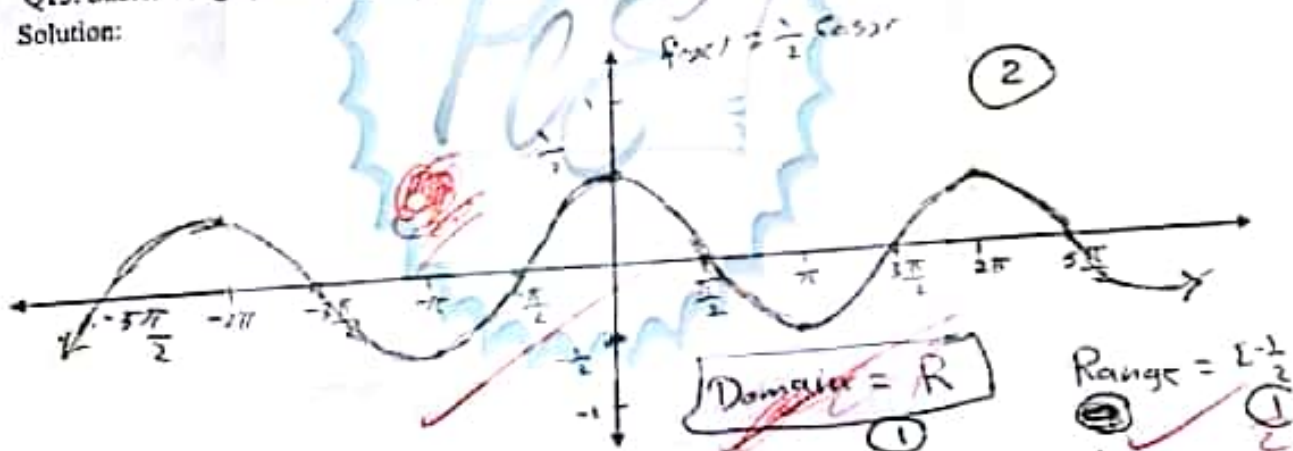
Q12. The set of values of x at which $f(x) = \frac{1}{|2x+1|-1}$, has discontinuities are:

- A) $\{0\}$ B) $\{-1\}$ C) $\{0, -1\}$ D) $\{0, 1\}$

Section 2: written questions.

Q13. Sketch the graph of $f(x) = \frac{1}{2} \cos(x)$, and then write out the domain and range. (5 marks)

Solution:



Q14. $\lim_{x \rightarrow -1} \frac{\sin(x+1)}{x^2-1} =$

⑤

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{\sin(x+1)}{(x-1)(x+1)} &= \lim_{x \rightarrow -1} \left(\frac{1}{x-1} \right) \left(\frac{\sin(x+1)}{x+1} \right) \\ &= \left(-\frac{1}{2} \right) (1) \\ &= -\frac{1}{2} \end{aligned}$$

or

(5 marks)

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{\sin(x+1)}{x^2-1} &= \lim_{x \rightarrow -1} \frac{1}{x-1} \cdot \lim_{x \rightarrow -1} \frac{\sin(x+1)}{x+1} \\ y &= x+1 \\ \text{as } x &\rightarrow -1 \Rightarrow y = 0 \\ &= -\frac{1}{2} \lim_{y \rightarrow 0} \frac{\sin y}{y} = -\frac{1}{2} (1) \\ &= -\frac{1}{2} \end{aligned}$$



مسئلة سنوات سابقة

المادة: حد كالكولاس ١/٣

Q1 :- If $f(x) = 3x - 2$ and $f \circ g(x) = 12x^2 - 6x + 16$
 $g(0)$?!

$$f \circ g(x) = f(g(x)) = 3(g(x)) - 2 = 12x^2 - 6x + 16$$

$$3(g(x)) = 12x^2 - 6x + 18$$

$$g(x) = 4x^2 - 2x + 6$$

$$g(0) = \underline{6} \quad (C)$$

Q2 :- Domain of $f(x) = \frac{\sqrt{9-x^2}}{f(x)} \cdot \frac{\sec x}{g(x)}$

Dom $h(x) = f(x) \cdot g(x)$ is $\text{Dom } f(x) \cap \text{Dom } g(x)$

Dom $f(x)$?!

$$(1) f(x) = \sqrt{9-x^2}, \quad \text{Dom } f(x) = [-3, 3]$$

$$9 - x^2 \geq 0$$

$$9 \geq x^2 \Rightarrow \pm 3 \geq x \Rightarrow -3 \leq x \leq 3$$

$$(2) \text{Dom } g(x) = \sec x = \mathbb{R} - \left\{ \frac{\pi}{2} + n\pi \right\}$$

n : Integer number.

$$\text{Dom } h(x) = [-3, 3] \cap \mathbb{R} - \left\{ \frac{\pi}{2} + n\pi \right\}$$

$$(3) \text{Dom } h(x) = [-3, 3] \sim \left\{ \frac{\pi}{2}, -\frac{\pi}{2} \right\} \quad (D)$$

Q3 :- $f(x) = 2x^3 + 5x + 3$, then the value of x such that $f^{-1}(x) = 0$

سؤال :- عند تعويض $f^{-1}(x)$ في $f(x)$ نصل إلى $f(f^{-1}(x)) = x$ وذلك لأن f^{-1} هو عكس f .

$$f(f^{-1}(x)) = x$$

$$f(0) = x, \quad 2(0) + 5(0) + 3 = x$$

$$\underline{x = 3}$$

(B)

Don't miss the change ... Be part of it

$$Q_4 :- \lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2}}{x^2} = \frac{0}{0} !!$$

$$\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2}}{x^2} \cdot \frac{1 + \sqrt{1 - x^2}}{1 + \sqrt{1 - x^2}}$$

$$\lim_{x \rightarrow 0} \frac{1 - (1 - x^2)}{x^2 + [1 + \sqrt{1 - x^2}]} = \lim_{x \rightarrow 0} \frac{x^2}{x^2} \cdot \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2}}{x^2} = \frac{1}{2} \quad (B)$$

$$Q_5 :- \lim_{x \rightarrow 0} \frac{\sin^2 x + 3x^2}{6x^2 + \cos^2 x}$$

$$\left. \begin{array}{l} \sin^2 x \in [0, 1] \\ \cos^2 x \in [0, 1] \end{array} \right\} \rightarrow 0 \text{ لذلك نعتبرهم } \text{ثوابت}$$

$$a = \sin^2 x$$

$$b = \cos^2 x$$

$$\lim_{x \rightarrow 0} \frac{a + 3x^2}{6x^2 + b} = \lim_{x \rightarrow 0} \frac{3x^2}{6x^2} = \frac{1}{2} \quad (A)$$

$$Q_6 :- f(x) = \sqrt[3]{x^3 - x} \text{ is}$$

$$f(-x) = \sqrt[3]{(-x)^3 - (-x)}$$

$$f(-x) = \sqrt[3]{-x^3 + x} = \sqrt[3]{-(x^3 - x)}$$

$$f(-x) = -\sqrt[3]{x^3 - x} = -f(x)$$

So $f(x)$ is odd function. (A)



سئلة سنوات سابقة

المادة:

Q7:- if $\lim_{x \rightarrow 0} f(x) = 1$, $\lim_{x \rightarrow 0} g(x) = -1$

then $\lim_{x \rightarrow 0} \frac{2f(x) - g(x)}{(f(x) + 7)^{2/3}} = ?!$

$$\lim_{x \rightarrow 0} \frac{2f(x) - g(x)}{(f(x) + 7)^{2/3}} = \lim_{x \rightarrow 0} \frac{2f(x)}{(f(x) + 7)^{2/3}} - \lim_{x \rightarrow 0} \frac{g(x)}{(f(x) + 7)^{2/3}}$$

$$= \frac{2 \lim_{x \rightarrow 0} f(x) - \lim_{x \rightarrow 0} g(x)}{\left(\lim_{x \rightarrow 0} (f(x) + 7) \right)^{2/3}}$$

$$= \frac{2 + 1}{(1 + 7)^{2/3}} = \frac{3}{\sqrt[3]{8^2}} = \frac{3}{\sqrt[3]{64}} = \sqrt[3]{\frac{3}{4}} \quad (A)$$

Q8:- $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 1} - 1}{x - 1} = \frac{\infty}{\infty} !!$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 \left(1 + \frac{1}{x^2}\right)} - 1}{x - 1} = \lim_{x \rightarrow \infty} \frac{|x| \sqrt{1 + \frac{1}{x^2}} - 1}{x - 1}$$

$$\lim_{x \rightarrow \infty} \frac{x \sqrt{1 + \frac{1}{x^2}} - 1}{x - 1} = \lim_{x \rightarrow \infty} \frac{x - 1}{x - 1} = 1$$

So at $y = 1$ there is H.A

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 1} - 1}{x - 1} = \lim_{x \rightarrow -\infty} \frac{|x| \sqrt{1 + \frac{1}{x^2}} - 1}{x - 1}$$

$$\lim_{x \rightarrow -\infty} \frac{-x - 1}{x - 1} = \lim_{x \rightarrow -\infty} \frac{-x}{x} = -2$$

So at $y = 1, -1$ there is only H.A (C)

Be part of it

سئلة سنوات سابقة

المادة:

Q9 :- $x=1$ and $x=-1$ are removable discontinuities. Verify.

$$\lim_{x \rightarrow 1^+} \frac{(x^3-1)^2}{(x^4-1)^2} = \frac{0}{0} !!$$

$$\lim_{x \rightarrow 1^+} \frac{[(x-1)(x^2+x+1)]^2}{[(x^2-1)(x^2+1)]^2}$$

$$\lim_{x \rightarrow 1^+} \frac{[(x-1)(x^2+x+1)]^2}{[(x-1)(x+1)(x^2+1)]^2} = \frac{(1^2+1+1)^2}{((1+1)(1^2+1))^2} = \frac{9}{16} \text{ not V.A}$$

$$\lim_{x \rightarrow -1^-} \frac{(x^3-1)^2}{(x^4-1)^2} = \frac{4}{0^+} = \infty \text{ So only } x=-1^- \text{ is V.A. (D)}$$

Q10 :- $\frac{x-1}{x+2} \neq 1$ (نرى أنه من الممكن الوصول لقيمة كالمية)

$$\lim_{x \rightarrow -2^+} \sqrt[3]{\frac{(x-1)}{(x+2)}} = \infty \Rightarrow R = \{1\} \text{ (D)}$$

Q11 :- The value K ?! $\frac{0}{0}$ is an indeterminate form.

$$\lim_{x \rightarrow 3} \sqrt{2x^2 - Kx} - 3 = 0$$

$$\sqrt{2(3)^2 - 3K} - 3 = 0$$

$$18 - 3K = 9$$

$$3K = 9, K = 3 \text{ (B)}$$

Q12 :- * نقاط عدم الاتصال إجمالاً ليست من ضمن مجال النهاية غير موجودة.

$$|2x+1| - 1 = 0$$

$$-(2x+1) = 1$$

نقاط عدم الاتصال

$$|2x+1| = 1$$

$$2x+1 = -1$$

$$x=0 \text{ و } x=-1$$

$$x=0$$

$$x=-1$$

(C)